

CLASS-8

LESSON-7 COMPARING QUANTITIES

(This PDF based on NCERT Book)

RECALLING RATIOS AND PERCENTAGES(अनुपातों तथा प्रतिशतों को याद करना)-

To turn a ratio into a percentage, you just need to follow two simple steps.

Think of a ratio like **1:4** (1 part juice, 4 parts water).

Step 1: Find the Total

First, add the numbers in the ratio together to see how many "total pieces" you have.

- **$1 + 4 = 5$ total pieces.**

Step 2: Divide and Multiply

Now, take the part you want to know about (the 1 part juice) and divide it by that total. Then, multiply by 100 to make it a percentage.

- $1 \div 5 = 0.20$
- $0.20 \times 100 = 20\%$

Another Easy Way: The Fraction Trick

You can also look at a ratio as a fraction and try to get the bottom number to **100**.

If you have the ratio **3:10** (3 parts sugar to 10 parts total):

1. Write it as a fraction: $\frac{3}{10}$
2. Ask: "What do I multiply 10 by to get 100?" (The answer is 10).
3. Do the same to the top: $3 \times 10 = 30$.
4. Your answer is **30%**.

Summary Table

Ratio (Part : Total)	As a Fraction	The Math	Percentage
1:2 (Half)	$\frac{1}{2}$	$1 \div 2 \times 100$	50%

Ratio (Part : Total)	As a Fraction	The Math	Percentage
1:4 (Quarter)	$\frac{1}{4}$	$1 \div 4 \times 100$	25%
1:5 (One-fifth)	$\frac{1}{5}$	$1 \div 5 \times 100$	20%
1:10 (One-tenth)	$\frac{1}{10}$	$1 \div 10 \times 100$	10%

To turn a ratio into a percentage, you can use this simple "Total-Parts" formula.

The Formula

$$\text{Percentage} = \frac{\text{Parts}}{\text{Total parts}} \times 100$$

FINDING DISCOUNTS(छूट ज्ञात करना)-

Now, imagine that same pizza costs **20**, but you have a **10% off** coupon.

The Formula:

$$\text{Savings} = \text{Price} \times \frac{\text{Discount}}{100}$$

The Steps:

1. **The Discount:** $10 \div 100 = 0.10$ (This is the "decimal" version).
2. **The Math:** $20 \times 0.10 = 2$
3. **The Result:** You save **2**.
4. **The Final Price:** $20 - 2 = 18$.

The "Easy Number" Cheat Sheet

If you don't want to do the big math, remember these "Magic Numbers" to multiply by:

If you want to find...	Multiply the price by:	Example (100 item)
5%	0.05	$100 \times 0.05 = 5$

If you want to find...	Multiply the price by:	Example (100 item)
10%	0.10	$100 \times 0.10 = 10$
25%	0.25	$100 \times 0.25 = 25$
50%	0.50	$100 \times 0.50 = 50$

Estimation in percentages-

The "10% Rule" (The Most Important Tool)

Almost all estimation starts with finding 10%. To find 10% of any number, just **move the decimal point one place to the left**.

- **10% of 500 is 50.**
- **10% of 85 is 8.5.**

1. How to Estimate 15% (The Tip Trick)

If you are at a restaurant and need to find a 15% tip:

1. **Find 10%:** (Move the decimal).
2. **Find 5%:** (Just cut your 10% number in half).
3. **Add them together.**

Example: Estimate 15% of **40**.

- 10% is 4.
- 5% is 2 (half of 4).
- **Total:** $4 + 2 = 6$.

2. How to Estimate 20%

This is even easier than 15%.

1. **Find 10%:** (Move the decimal).
2. **Double it.**

Example: Estimate 20% of **70**.

- 10% is 7.
- $7 \times 2 = 14$.

3. The "Rounding" Strategy

If a number is messy, round it to the nearest "clean" number (ending in 0) before you start.

- **To find 25% of 39:** * Round 39 to **40**.
 - 25% is the same as **one-quarter** (1/4).
 - $40 \div 4 = 10$. (The real answer is 9.75, so 10 is a great estimate!)

SALES TAX / VALUE ADDED TAX / GOODS AND SERVICE TAX (विक्री कर/ मूल्य वर्धित कर/ वस्तु एवं सेवा कर)-

- **Sales Tax:** An extra amount added to the price of an item at the **final** checkout.
- **VAT / GST:** A tax that is added at **every step** of making a product (from the farmer to the factory to the store).

The Result for You: Whether it's VAT, GST, or Sales Tax, the price you see on the tag (or at the register) usually includes this extra percentage that goes to the government to pay for roads, schools, and hospitals.

2. How to Calculate It (The Formula)

Since these are all **percentages**, you use the same math we used for discounts—but instead of subtracting, you **add** the amount.

The Formula:

$$\text{Tax Amount} = \text{Price} \times \frac{\text{TaxRate}}{100}$$

3. A Simple Example (The 100 Phone Case)

Imagine you buy a phone case for **100** and the tax (GST/VAT) is **18%**.

1. **Find the Tax:** $18 \div 100 = 0.18$
2. **Multiply:** $100 \times 0.18 = 18$ (This is the tax money).
3. **Add it up:** $100 + 18 = 118$ (This is your final "Total" price).

COMPOUND INTEREST(चक्रवृद्धि ब्याज)-

The Formula

$$A = P \left(1 + \frac{R}{100} \right)^n$$

A (Amount): This is the **Final Total**. It includes your original money plus all the interest you earned.

- **P (Principal):** The **Starting Money**. The amount you originally deposited or borrowed.
- **R (Rate):** The **Interest Percentage** (e.g., if it's 5%, you just put "5" in the formula).
- **n (Time):** The **Number of Years** (or compounding periods) the money stays in the account.

A Real-Life Example

Imagine you put **5,000** into a savings account at **8%** interest for **3 years**.

1. Fill in the formula:

$$A = 5000 \left(1 + \frac{8}{100} \right)^3$$

2. Solve the bracket first:

- $8 \div 100 = 0.08$
- $1 + 0.08 = 1.08$

3. Apply the power ($1.08 \times 1.08 \times 1.08$):

- $(1.08)^3 \approx 1.2597$

4. Multiply by the starting money:

- $5000 \times 1.2597 = 6,298.50$

How to find the "Interest Only"

If a question asks you "How much interest did you earn?" (Compound Interest or **C.I.**), you just subtract your starting money from your final total:

$$\text{C.I.} = A - P$$

In our example: $6,298.50 - 5,000 = 1,298.50$.

DEDUCING A FORMULA FOR COMPOUND INTEREST(चक्रवृद्धि ब्याज का सूत्र निकालना):-

To deduce the formula $A = P \left(1 + \frac{R}{100} \right)^n$, we simply follow the money year by year.

The goal is to see how your money grows when the interest from one year is added to your total to earn **even more** interest the next year.

Year 1: The Start

You start with your Principal (P). At the end of the first year, the bank gives you interest based on the rate (R).

- **Interest for Year 1** $= P \times \frac{R}{100}$
- **Total Amount (A_1)** $= P + (P \times \frac{R}{100})$
- **Factor out the P:** $A_1 = P(1 + \frac{R}{100})$

Year 2: Compounding Begins

This is the "Compound" part. In Year 2, your starting money isn't just P; it is the **total amount from Year 1 (A_1)**.

- **Total Amount (A_2)** $= A_1 + (\text{Interest on } A_1)$
- **Simplified:** $A_2 = A_1 \left(1 + \frac{R}{100}\right)$
- **Substitute A_1 :** Since we know $A_1 = P(1 + \frac{R}{100})$, we plug that in:
- **Result:** $A_2 = [P(1 + \frac{R}{100})] \times (1 + \frac{R}{100})$
- **Final Year 2:** $A_2 = P(1 + \frac{R}{100})^2$

Year 3: The Pattern Continues

Following the same logic, for Year 3, we take the Year 2 total and multiply it by the growth factor again.

- **Total Amount (A_3)** $= A_2 \times (1 + \frac{R}{100})$
- **Result:** $A_3 = P(1 + \frac{R}{100})^3$

The General Formula (Year n)

Notice that the **power** (the small number at the top) always matches the **number of years** that have passed.

- After 1 year → Power is 1
- After 2 years → Power is 2
- After 3 years → Power is 3

Therefore, for **n** number of years, the total amount **A** is:

$$A = P(1 + \frac{R}{100})^n$$

APPLICATION OF COMPOUND INTEREST FORMULA(चक्रवृद्धि ब्याज का सूत्र का

अनुप्रयोग)-

1. Population Growth

Towns and cities grow just like money. If a city has a certain number of people and grows by a percentage each year, the new residents also have children, causing the population to "compound."

- **P:** The current population.
- **R:** The birth/growth rate.
- **n:** Number of years.

2. Depreciation (Losing Value)

This is "Compound Interest in reverse." When you buy a car or a mobile phone, it loses value every year. This is called **Depreciation**.

- **The Tweak:** We change the plus sign to a minus because the value is going **down**.
- **Formula:** $A = P(1 - \frac{R}{100})^n$

Example: If you buy a phone for 50,000 and it loses 10% value each year, you use this formula to find its "Resale Value" after 3 years.

3. Bacteria and Biology

In a lab, bacteria cells split and double. If you start with 100 cells and they grow at a steady rate, the formula tells you how many millions you will have in a few days. This is why infections can spread so quickly!

4. Inflation (Rising Prices)

Inflation is the reason a chocolate bar cost much less 10 years ago than it does today. It is the "Compound Interest" of prices.

- If inflation is 5%, it means the price of goods is "compounding" upwards every year. The formula helps economists predict what a loaf of bread will cost in the year 2030.

Summary Table: When to use the formula

Scenario	Is it growing or shrinking?	Formula Sign
Savings/Investment	Growing	+ (Plus)
Population Growth	Growing	+ (Plus)

Scenario	Is it growing or shrinking?	Formula Sign
Inflation (Prices)	Growing	+ (Plus)
Depreciation (Car/Tech)	Shrinking	- (Minus)
Radioactive Decay	Shrinking	- (Minus)